Chapter 17
Durability: oxidation, corrosion and degradation



Corrosion. (Image courtesy of Norbert Wodhnl © Norbert Wodhnl)

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17.1 Introduction and synopsis

The Gospel according to St Matthew reminds us that we live in a world in which 'moth and rust doth corrupt'—a world of corrosion, degradation and decay (cover picture). Not a happy thought with which to start a chapter. But by understanding them we can, to a degree, control them. This chapter describes ways in which materials degrade or corrode, how they are measured and what can be done to slow them down.

Start with the first: the ways in which materials degrade. There are many. There is the damaging effect of radiation and heat, not in changing the properties in a reversible way as in the last chapter, but in causing chemical changes that destroy them irreversibly. There is attack by aggressive chemicals: acids, alkalis and, in the case of polymers, organic solvents. And there is aqueous corrosion, even in pure fresh water, which, provided oxygen is dissolved in it, can be remarkably aggressive.

Because many different properties are involved here, the exploration of the science is split into two, one delving into oxidation, flammability and photo-degradation, the other into corrosion by liquids. Each section ends with a description of ways in which degradation can be inhibited.

17.2 Oxidation, flammability and photo-degradation

The most stable state of most elements is as an oxide. For this reason the earth's crust is almost entirely made of simple or complex oxides: silicates like granite, aluminates like basalt, carbonates like limestone. Techniques of thermochemistry, electrochemistry and synthesis allows these to be refined into the materials we use in engineering, but they are not, in general, oxides. From the moment they are made to re-oxidize, some extremely slowly, others more quickly; and the hotter they are, the faster it happens. For safe high-temperature design we need to understand rates of oxidation.

Definition and measurement

Oxidation rates of metals are measured in the way sketched in Figure 17.1: a thin sheet of the material (to give a lot of surface) is held at temperature T for an increasing time t, and the gain or loss in weight, Δm, is measured. If the oxide adheres to the material the sample gains weight in a way that is either linear ($\Delta m \propto t$) or parabolic ($\Delta m \propto t^{1/2}$) in time t; if instead the oxide is volatile, the sample loses weight linearly with time ($\Delta m \sim -t$), as in Figure 17.2.

Polymers, too, oxidize, but in a more spectacular way.

Flammability

Ceramics and glasses do not burn. Metals, if dispersed as a fine powder, are potentially combustible, but in bulk flammability is not normally an issue, even for magnesium. Most polymers, on the other hand, are inherently flammable,